Two-FPGA Test Setup: Board-to-Board Link Pinout

conv_filter_chip post-silicon validation (PYNQ-Z2 x 2)

The tester board drives the 14 chip inputs and samples the 11 chip outputs of the DUT board over the Raspberry Pi 40-pin header of each PYNQ-Z2. Both boards use the same header pin numbers: connect pin N on the tester to pin N on the DUT board with a female-female jumper.

- 25 signal wires (every GPIO pin except 7, 26 and 29) plus ground wires: use at least pins 9 and 14 (next to the clock on pin 11), 25 and 39; all eight ground pins if possible.
- **Never connect pins 1, 2, 4 or 17** (3V3/5V). Each board is powered from its own USB.
- Leave pins 7, 26 and 29 unconnected. Leave PMOD A unplugged on both boards: six of its pins are shared with this header.
- Keep jumpers short (10-15 cm or less), especially the clock on pin 11. Load the DUT bitstream (make program-dut) only on the DUT board.
- Link defaults: 8.33 MHz (H=3), DUT outputs sampled at the next rising edge (S=0). Inputs change on the link clock's falling edge. Check margins with jtb_sweep_sample.

Signals

Signal	Direction	RPi pin	BCM GPIO	FPGA pin	Notes
chip_clk	tester → DUT	11	GPIO17	U7	forwarded link clock, clock-capable
chip_kernel_valid	tester → DUT	8	GPIO14	V6	
chip_normalization_valid	tester → DUT	10	GPIO15	Y6	
chip_pixel_valid	tester → DUT	12	GPIO18	C20	
chip_reset_n	tester → DUT	3	GPIO2	W18	JA-shared, pull-up (non-critical)
chip_test_mode	tester → DUT	5	GPIO3	W19	JA-shared, pull-up (non-critical)
chip_datain[0]	tester → DUT	15	GPIO22	U8	
chip_datain[1]	tester → DUT	16	GPIO23	W6	
chip_datain[2]	tester → DUT	18	GPIO24	Y7	
chip_datain[3]	tester → DUT	19	GPIO10	V8	
chip_datain[4]	tester → DUT	21	GPIO9	V10	
chip_datain[5]	tester → DUT	22	GPIO25	F20	
chip_datain[6]	tester → DUT	23	GPIO11	W10	
chip_datain[7]	tester → DUT	24	GPIO8	F19	
chip_dataout[0]	DUT → tester	31	GPIO6	U18	JA-shared (non-critical)
chip_dataout[1]	DUT → tester	32	GPIO12	B20	
chip_dataout[2]	DUT → tester	33	GPIO13	W8	
chip_dataout[3]	DUT → tester	35	GPIO19	Y8	
chip_dataout[4]	DUT → tester	36	GPIO16	B19	
chip_dataout[5]	DUT → tester	37	GPIO26	W9	
chip_dataout[6]	DUT → tester	38	GPIO20	A20	
chip_dataout[7]	DUT → tester	40	GPIO21	Y9	
chip_output_valid	DUT → tester	13	GPIO27	V7	
chip_status[0]	DUT → tester	27	GPIO0	Y16	JA-shared, pull-up (non-critical)
chip_status[1]	DUT → tester	28	GPIO1	Y17	JA-shared, pull-up (non-critical)

FPGA pin is the xc7z020 package pin, the same on both boards (official TUL PYNQ-Z2 v1.0.xdc, rpio_NN_r). Pin 13 (GPIO27, V7) is missing from that file and from Vivado's board files; it was verified with the I/O lab.

Pin placement. Pins 3, 5, 7, 26-29 and 31 share their net with PMOD JA (3, 5, 27, 28 also have pull-ups). As receivers at 50 MHz their falling edges are ˜4.7 ns (˜10 ns with the pull-up) slower than rising, vs 1-3 ns on plain pins. So all critical signals use plain pins, the pulled-up pins carry only reset_n, test_mode and status, and dataout[0] (31) is the only JA-shared data bit. Buses run in bit order: controls on 8-13, datain[0..7] on 15-24, dataout[0..7] on 31-40.

Header wiring checklist (both boards, same pin numbers)

Pin	Connect	Signal	Done	Pin	Connect	Signal	Done
1	**NO**	3V3 (do not connect)		2	**NO**	5V (do not connect)	
3	yes	chip_reset_n	□	4	**NO**	5V (do not connect)	
5	yes	chip_test_mode	□	6	yes	GND	□
7	no	unused		8	yes	chip_kernel_valid	□
9	yes	GND	□	10	yes	chip_normalization_valid	□
11	yes	chip_clk	□	12	yes	chip_pixel_valid	□
13	yes	chip_output_valid	□	14	yes	GND	□
15	yes	chip_datain[0]	□	16	yes	chip_datain[1]	□
17	**NO**	3V3 (do not connect)		18	yes	chip_datain[2]	□
19	yes	chip_datain[3]	□	20	yes	GND	□
21	yes	chip_datain[4]	□	22	yes	chip_datain[5]	□
23	yes	chip_datain[6]	□	24	yes	chip_datain[7]	□
25	yes	GND	□	26	no	unused	
27	yes	chip_status[0]	□	28	yes	chip_status[1]	□
29	no	unused		30	yes	GND	□
31	yes	chip_dataout[0]	□	32	yes	chip_dataout[1]	□
33	yes	chip_dataout[2]	□	34	yes	GND	□
35	yes	chip_dataout[3]	□	36	yes	chip_dataout[4]	□
37	yes	chip_dataout[5]	□	38	yes	chip_dataout[6]	□
39	yes	GND	□	40	yes	chip_dataout[7]	□

Odd pins are on one row of the header and even pins on the other; pin 1 is marked on the board silkscreen. Check pin 1 and a ground pin against the PYNQ-Z2 manual before powering up.

DUT-board inputs have pull-downs, so an unwired or idle link holds the DUT in reset, except on the pulled-up pins 3 and 5 (chip_reset_n, chip_test_mode), which float high while the tester is unprogrammed; harmless without a forwarded clock.